

3.1 Newton's Laws of Motion

Question Paper

Course	CIE A Level Physics
Section	3. Dynamics
Topic	3.1 Newton's Laws of Motion
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A cyclist is riding at a steady speed on a level road

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

- A. the force exerted by the cyclist on the pedals
- B. the total air resistance and friction force
- C. the tension in the cycle chain
- D. the forward push of the road on the back wheel

[1 mark]

Question 2

A resultant force causes a body to accelerate.

What is equal to the resultant force?

- A. the acceleration of the body per unit mass
- B. the change in kinetic energy of the body per unit time
- C. the change in momentum of the body per unit time
- D. the change in velocity of the body per unit time

[1 mark]

Question 3

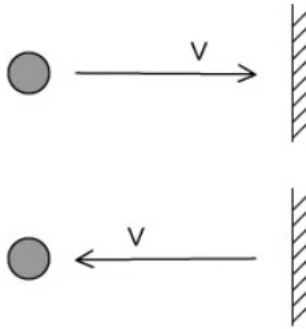
What is meant by the mass and by the weight of an object on the Earth?

	Mass	Weight
A	its momentum divided by its velocity	the work done in lifting it one metre
B	the gravitational force on it	the property that resists its acceleration
C	the property that resists its acceleration	the pull of the Earth on it
D	the pull of the Earth on it	its mass divided by the acceleration of free fall

[1 mark]

Question 4

An object travelling with velocity v strikes a wall and rebounds as shown



Which property of the object is **not** conserved?

- A. momentum
- B. mass
- C. kinetic energy
- D. speed

[1 mark]

Question 5

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a .

Which quantities must be in the same direction?

- A. a and v only
- B. v , a and F
- C. v and F only
- D. F and a only

[1 mark]

Question 6

An egg is dropped from the top of a three-storey building. It falls through air until it reaches the ground

What remains constant throughout the fall?

- A. acceleration of the egg
- B. weight of the egg
- C. velocity of the egg
- D. air resistance on the egg

[1 mark]

Question 7

Which is **not** one of Newton's laws of motion?

- A. if body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A
- B. the rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force
- C. the total momentum of a system of interacting bodies remains constant, providing no external force acts
- D. a body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force

[1 mark]

Question 8

Question 8

The symbol g represents the acceleration of free fall

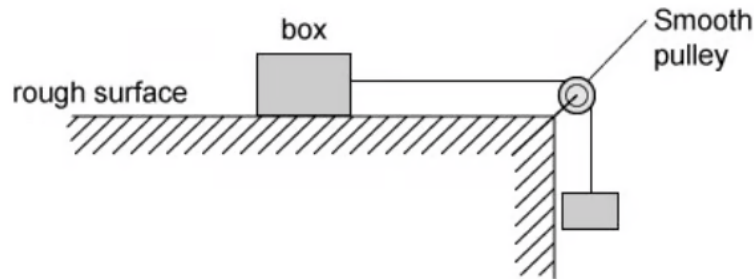
Which of these statements is correct?

- A. g is gravity
- B. g is reduced by air resistance
- C. g is the weight of an object
- D. g is the ratio weight / mass

[1 mark]

Question 9

A box rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a mass at its other end.



When the box is released, what forces are acting on it?

- A. weight only
- B. friction and weight
- C. friction, weight and reaction
- D. tension, friction, weight and reaction

[1 mark]

Question 10

As a car accelerates uniformly, its momentum is measured at regular intervals

A graph of the momentum of the car is plotted against time

What is evaluated by finding the gradient of the graph at a particular time?

- A. the acceleration of the car
- B. the resultant force on the car
- C. the kinetic energy of the car
- D. the velocity of the car

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- The question tells you a key statement about Newton's Third Law (that the force is equal and opposite)
- But there are two key pieces of information about the forces in Newton's Third Law you will need to answer this question
- Firstly, the force acts between the same two objects. So, if one force is from the back wheel on the road, the other **must** be from the road on the back wheel
- Secondly, the two forces must be the same kind of force. In this particular case, the force on the wheel on the road is a friction force, so the force of the road on the wheel must be a friction force, too

A, B & C are incorrect because the force must act between the same two objects (the back wheel and the road), and be the same force (frictional) whereas these options all reference different bodies and different forces

2

The correct answer is **C** because:

- The resultant force is equal to the rate of change of momentum, which is the same as saying the change in momentum of the body per unit time
- This is given by Newton's Second Law:

$$\circ F = ma = \frac{m \Delta v}{t} = \frac{\Delta p}{t}$$

A is incorrect because the acceleration of the body per unit mass means $\frac{m}{a}$ which does not equate to force

B is incorrect because the change in kinetic energy of the body per unit time (rate of change of energy) gives the power

D is incorrect because the change in velocity of the body per unit time (rate of change of velocity) gives acceleration

3

The correct answer is **C** because:

- **Mass** is the amount of matter contained in a body.
- It is a constant quantity and does not change with a change of position or location
- Consider the equation $F = ma$. Given a constant force, F , a smaller value of m will yield a larger acceleration, a , because the two sides of the equation must always be equal. Thus, larger masses produce smaller accelerations.
- Or another way to put it, the higher the mass of the body, the more resist to uniform motion (acceleration) it is
- **Weight** is the force exerted by a body when it is in a gravitational field
- The weight of the body depends upon the gravitational field and acceleration due to gravity at that location, in this case, the pull of the Earth on the body
- But on Mars, for example, due to the smaller size and mass of Mars compared to the Earth, the acceleration due to gravity is lower, so whilst the mass of a body would remain the same, the weight of the body would be less

A is incorrect because the definition of mass is correct, but the definition of weight indicates a force working against gravity

B is incorrect because these definitions are correct but are the wrong way around

D is incorrect because both of these are definitions to describe the weight

4

The correct answer is **A** because:

- Momentum is a vector quantity
- Initial momentum = mv , final momentum = $-mv$
- Even though the magnitude of the momentum remains the same, the direction of the velocity changes
- The momentum of the object is not conserved but the momentum of the **whole system** should be conserved. The question asks which 'property of the object' is not conserved, not the whole system, so this is an important distinction

B, C & D are incorrect because these are scalar quantities, and the magnitudes of m and v stay the same so kinetic energy, speed and mass are all conserved

5

The correct answer is **D** because:

- According to Newton's Second Law, a force causes an object to accelerate in the same direction it is applied

A, B & C are incorrect because Newton's First Law states that an object will remain at rest or at the same constant speed unless a resultant force acts upon it. This means a force could be applied in the same direction as an object's motion, causing it to accelerate, or an opposing force could be applied, to cause it to decelerate. Therefore, the direction of the velocity does not have to be the same direction as the force or acceleration

6

The correct answer is **B** because:

- Weight = mass $\times$ acceleration due to gravity (g)
- Both the mass of the egg and the value of g stay constant during the fall

A is incorrect because the acceleration decreases as the magnitude of the air resistance increases

C is incorrect because the velocity increases until it reaches its terminal velocity

D is incorrect because air resistance increases as the velocity of the egg increases

7

The correct answer is **C** because:

- This is the principle of conservation of momentum which is a direct consequence of Newton's Third Law, but it does not describe it
- Newton's Third Law states that for every force there is an equal and opposite force, from this we can derive conservation of momentum, which tells us that a change in momentum in one of two objects colliding is equal and opposite to the change in

momentum in the other colliding object

A is incorrect because this is Newton's Third Law

B is incorrect because this is Newton's Second Law, $F = ma = \frac{m \Delta v}{t} = \frac{\Delta p}{t}$

D is incorrect because this is Newton's First Law

8

The correct answer is **D** because:

- Weight = mass $\times$ acceleration due to gravity (g)
- So, $g = \text{weight} / \text{mass}$

A & C are incorrect because gravity/weight is a force, whereas g is the acceleration

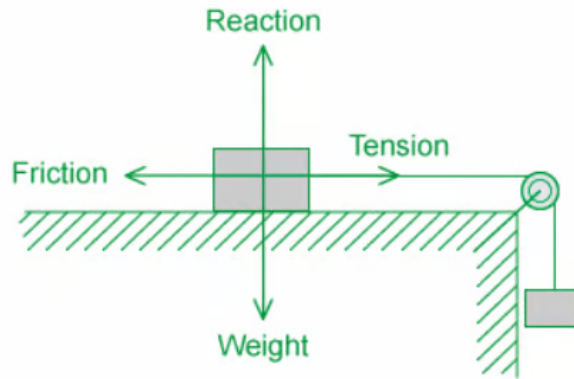
B is incorrect because the value of g cannot be changed by air resistance as it is constant all over the Earth's surface

9

The correct answer is **D** because:

- The weight of the mass below the pulley induces a tension in the wire which pulls the box to the right
- The force of friction opposes the motion and acts to the left, the magnitude of the friction is less than the tension it is represented by a shorter arrow
- The weight of the box acts downwards, and since the box is on a table there is an upward reaction force between the table and the box

- The force of friction opposes the motion and acts to the left, the magnitude of the friction is less than the tension it is represented by a shorter arrow
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10

The correct answer is **B** because:

- The gradient of a momentum–time graph is equal to
 - $\frac{\Delta y}{\Delta x} = \frac{\text{change in momentum}}{\text{time}} = \frac{\Delta p}{t}$
- Momentum is equal to $\Delta p = m \Delta v$
 - $\frac{\Delta p}{t} = \frac{m \Delta v}{t}$
- Acceleration is equal to $a = \frac{\Delta v}{t}$
 - $\frac{m \Delta v}{t} = ma = F$
- Therefore, the gradient is equal to the resultant force on the car

